Run report

2026-10-30T161244Z

Configuration

Parameter	Value	Parameter	Value
batch_size	1,024	n_samples	500,000
dim	2	dataset	gaussian
seed	1,766,738,368	data_seed	42
hidden_units	64	n_layers	2
n_epochs	100	n_disc_steps	5
lr_g	3.000×10^{-5}	lr_d	1.000×10^{-4}
lambda_dispersion	0.01500	mmd_subsample	16,384
mmd_test	5.388×10^{-5}	best_epoch	96
compile_cache_warm	True		
gaussian_params	dim=2, mu_gen=[0, 1], mu_true=[0.2, 0.8], cov_gen=[[1, -0.9], [-0.9, 2.25]], cov_true=[[0.81, -0.702], [-0.702, 1.69]], cov_detector=[[0.25, 0], [0, 0.64]]		
mmd_sigmas_detector	$1.460 \times (\frac{1}{2},\ \frac{1}{\sqrt{2}},\ 1,\ \sqrt{2},\ 2)$		
mmd_sigmas_particle	$1.217 \times (\frac{1}{2},\ \frac{1}{\sqrt{2}},\ 1,\ \sqrt{2},\ 2)$		

Timing

Phase	Seconds	Share (%)	Detail
Deconvolve			
data	0.140	0.6	gaussian cache hit – train
train	18.068	76.2	train
transfer	6.176		train
compile	2.647		train
epochs	6.353		train
select	2.561		train
particle_mmd	0.482	2.0	train
save	0.159	0.7	train
evaluate	4.858	20.5	train
total	23.707	100.0	
IBU baseline			
parse_config	0.002	0.0	ibu
data	0.155	2.7	gaussian cache hit – ibu
ibu	0.165	2.8	niter=10 – ibu
evaluate	5.496	94.5	2 variables – ibu
total	5.817	100.0	
OmniFold baseline			
parse_config	0.001	0.0	omnifold
data	0.152	0.0	gaussian cache hit – omnifold
omnifold	354.347	98.3	niter=3, epochs=50 – omnifold
init	3.768		DataLoaders and the two MLPs – omnifold
unfold	310.973		MultiFold.Unfold – omnifold
iter1_step1	79.649		detector-level reweighting – omnifold
iter2_step1	44.113		detector-level reweighting – omnifold
iter3_step1	33.678		detector-level reweighting – omnifold
iter1_step2	47.586		particle-level reweighting – omnifold
iter2_step2	35.865		particle-level reweighting – omnifold
iter3_step2	70.049		particle-level reweighting – omnifold
reweight	1.545		evaluating weights on z_target – omnifold
evaluate	5.810	1.6	2 variables – omnifold
total	360.311	100.0	
Common			
parse_config	0.001	0.0	plot
data	0.143	0.6	gaussian cache hit – plot
load	1.449	6.3	plot
plots	21.271	93.0	plot
total	22.863	100.0	

Detector-level metrics

Distance to Data. Improvements are relative to Simulation.

Wasserstein Distance [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	207.1	**7.804**	**96.2**	11.86	94.3	14.32	93.1
dim_1	230.1	27.45	88.1	**12.66**	**94.5**	17.47	92.4

JS divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	6.676	0.2937	95.6	**0.2904**	**95.6**	0.3090	95.4
dim_1	5.198	0.6584	87.3	**0.2787**	**94.6**	0.3121	94.0

VLC divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	26.17	1.150	95.6	**1.139**	**95.6**	1.212	95.4
dim_1	20.22	2.506	87.6	**1.072**	**94.7**	1.199	94.1

Sliced Wasserstein, joint [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
All	162.5	61.78	62.0	**10.47**	**93.6**	13.02	92.0

Mean 1D distance over random directions, in Data standard deviations. IBU: product of its per-observable weights.

Particle-level metrics

Distance to Truth. Improvements are relative to Generation.

Wasserstein [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	204.8	20.17	90.2	**11.19**	**94.5**	13.62	93.3
dim_1	242.0	48.17	80.1	**15.66**	**93.5**	22.57	90.7

JS divergence [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	8.734	1.930	77.9	**0.2602**	**97.0**	0.2862	96.7
dim_1	8.100	2.746	66.1	**0.2567**	**96.8**	0.3114	96.2

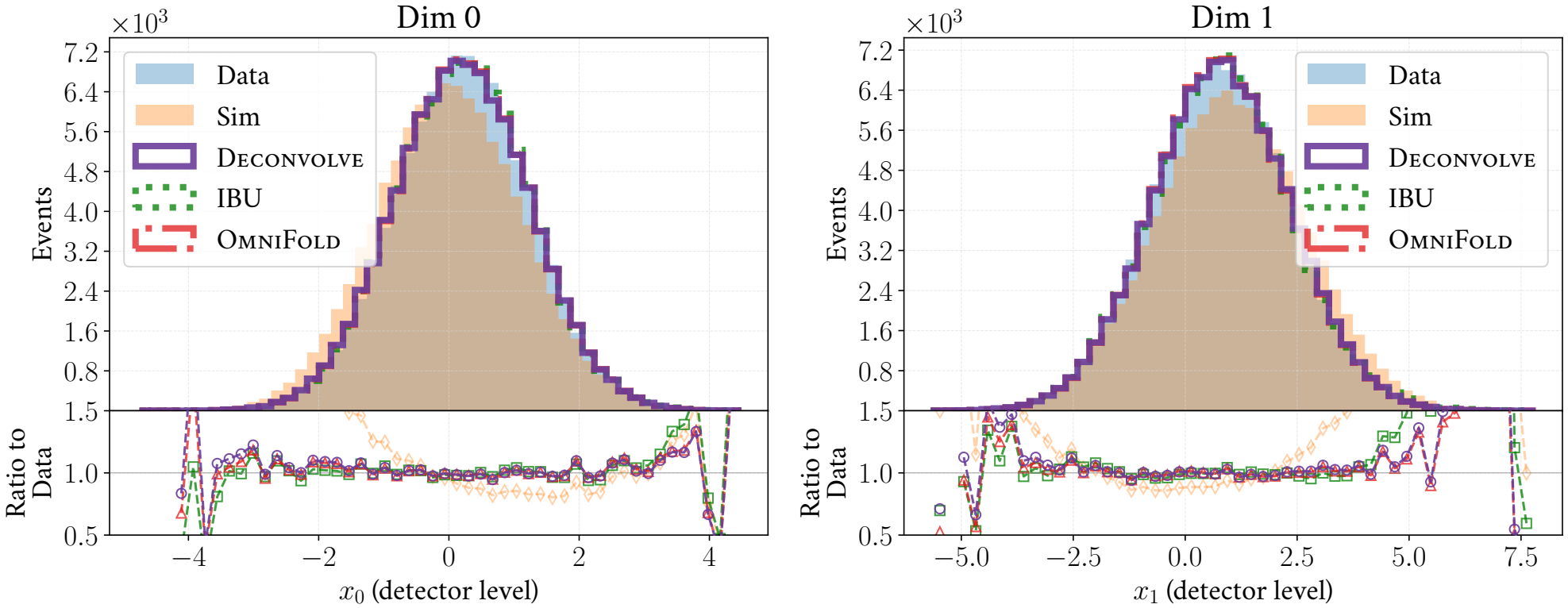
VLC divergence [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	34.05	7.635	77.6	**1.009**	**97.0**	1.108	96.7
dim_1	31.35	10.78	65.6	**1.009**	**96.8**	1.220	96.1

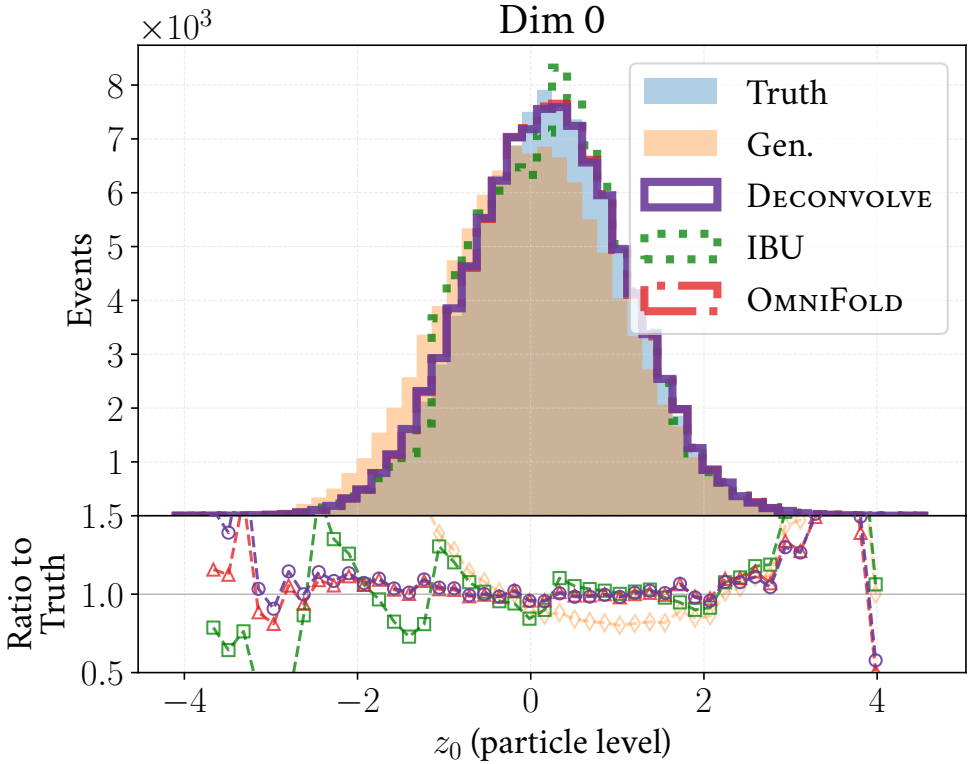
Sliced Wasserstein, joint [$\times 10^3$]

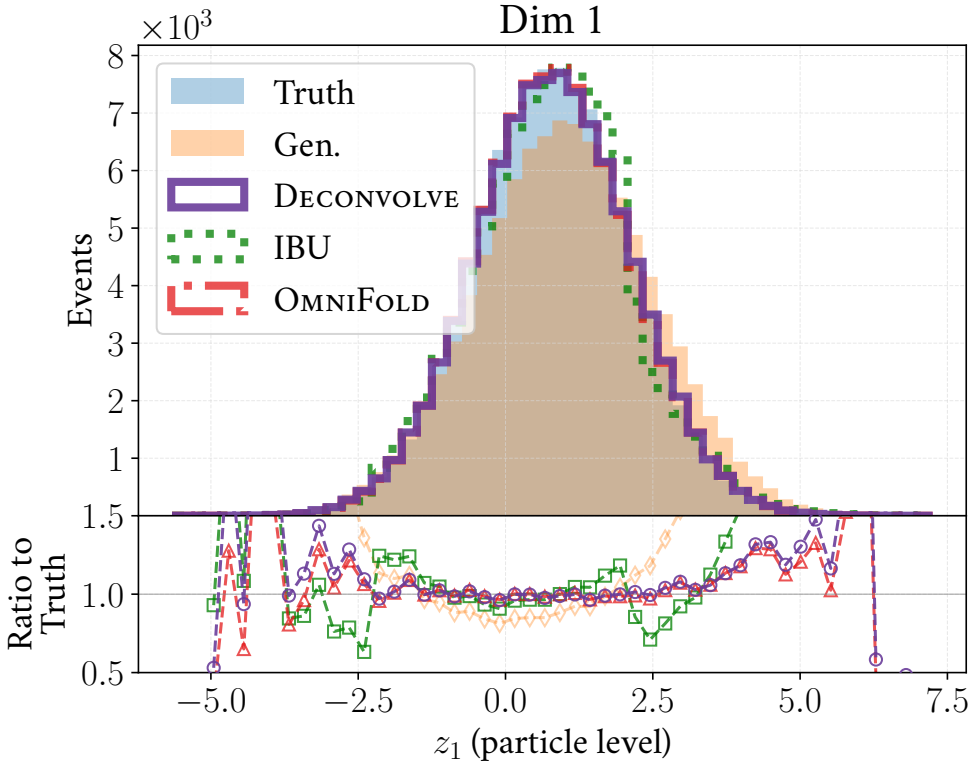
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
All	193.1	76.13	60.6	**13.38**	**93.1**	16.83	91.3

Mean 1D distance over random directions, in Truth standard deviations. IBU: product of its per-observable weights.


Detector Level
Dim 0
Dim 1
×10^3
Events
Ratio to Data
Data
Sim
Deconvolve
IBU
OmniFold
x_0 (detector level)
x_1 (detector level)

Particle Level





Training diagnostics

Training losses



Selection criteria

